

What is a swerve drivetrain?

A swerve drivetrain is an advanced drivetrain system where each individual wheel can independently drive and rotate a full 360 degrees. This allows the robot to move in a holonomic motion (forwards, backwards, diagonally, or sideways) and rotate in place at the same time. A standard swerve drivetrain typically consists of four independent modules placed at the corners of the drivetrain. Unlike the more common mecanum drivetrains, the swerve drivetrain usually uses treaded traction wheels, or custom wheels giving the robot full pushing power without the risk of being easily pushed around by opponents.

What are the different types?

There are two different types of swerve drivetrains which are distinguished by their module architecture. The first type is the coaxial swerve. This drivetrain is standard for competitive robotics like First Tech Challenge (FTC). The wheels are driven by a motor, however the steering is done by a servo, due to the motor constraints in FTC. The second type is the differential swerve. The differential swerve is much more complex and powerful than the coaxial swerve, as both the steering and wheels are driven by motors. Both motors contribute to the steering, and the wheel driving as they are placed in a gearbox shown in the image below:



What are the Pros and Cons of the Differential Swerve?

The differential swerve drive has a very complex design, as it requires both motors per module to contribute to driving the wheel and steering it. The differential swerve drive can be extremely compact and can reduce the overall size of the robot even while maintaining power in the wheels. Differential swerve drivetrains are usually used in FRC. This is because it requires 8 motors to run 4 separate modules. However some FTC teams have built a differential swerve drive by only utilizing 2 modules in opposite corners. In the other corners, they would place odometry dead wheels for localization. This is good because it allows the robot to stay compact.

However this reduces the overall push that the robot can provide, weakening it, this retracts the whole purpose of the differential swerve drivetrain.

What are the Pros and Cons of the Coaxial Swerve?

The Coaxial swerve drivetrain on the other hand has a much simpler design. This is because it only utilizes one motor and one servo per module. This allows us to use the other 4 motors for other subsystems in the robot. It also still has a stronger push than a mecanum drivetrain as it uses traction wheels even though it doesn't use two motors per module like the differential swerve. However, to make this design compact, we would have to use GoBuilda bare motors which would require it to gear it down to at least 500 rpm from 6,000 rpm. This increases the overall size of the drivetrain and robot.

Which one is better?

For our learning and FTC requirements, I would choose the coaxial swerve drivetrain. Although it will increase the overall size of the robot, if we design it in a parallel plate drive train, it wouldn't make much of a difference if it would still fit in the pod.